

Lecture 7: Asynchronous programming

Presentations students, introduction to async

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DevLab Rust 2025

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github.com/wvhulle/rust-course-ghent

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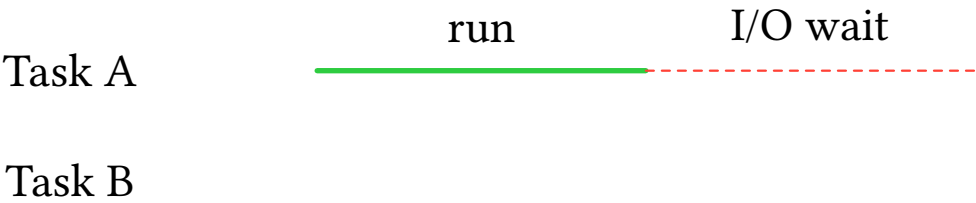
2.1. Introduction

Multiple tasks execute concurrently by running until they would block on I/O, then yielding to another ready task.



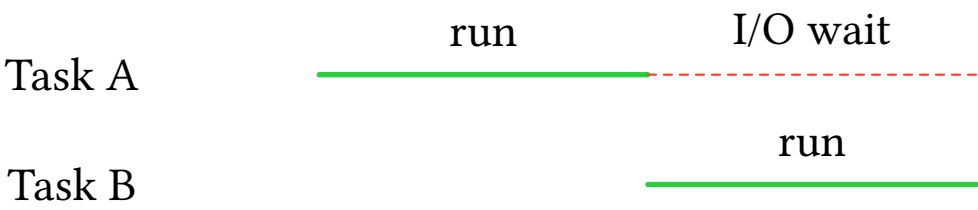
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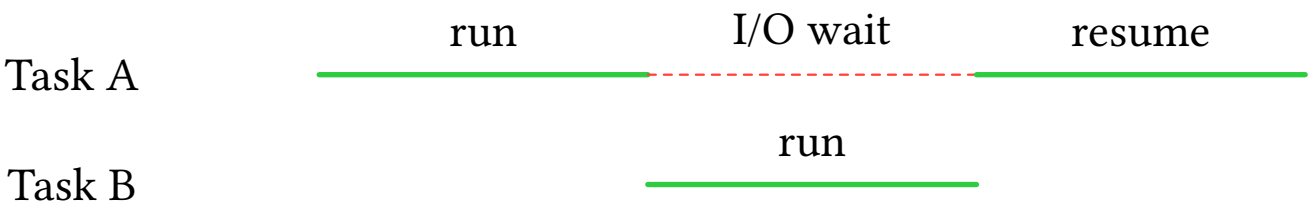


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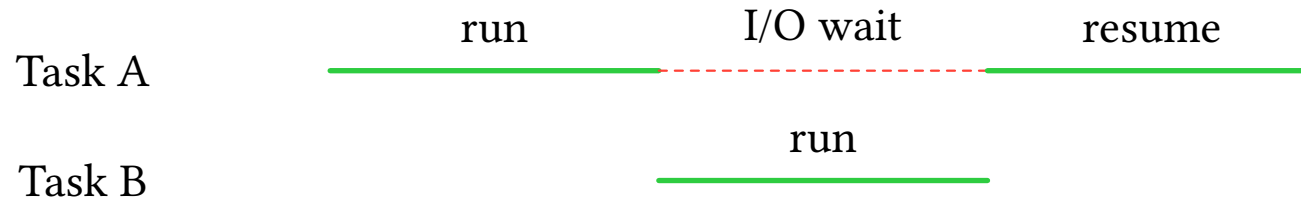


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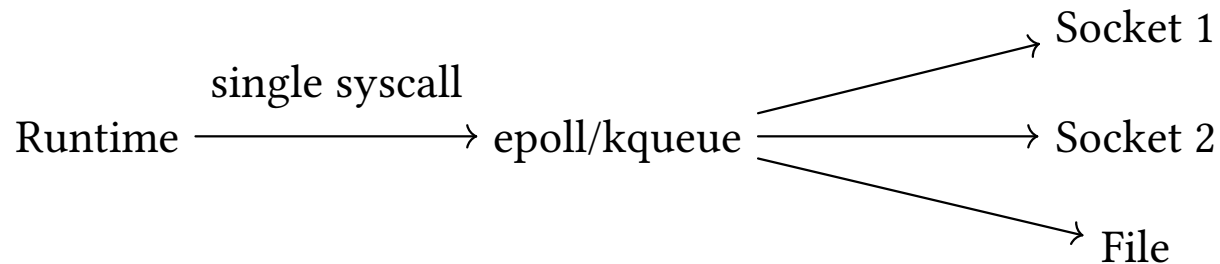


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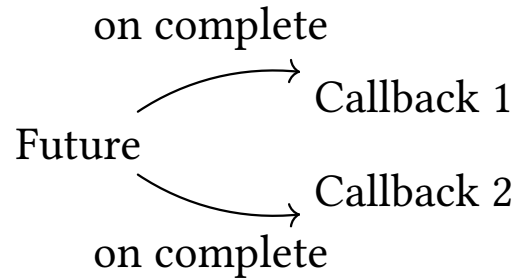


Per-task overhead is minimal because operating systems provide primitives (epoll, kqueue, IOCP) that monitor many I/O operations with a single system call.



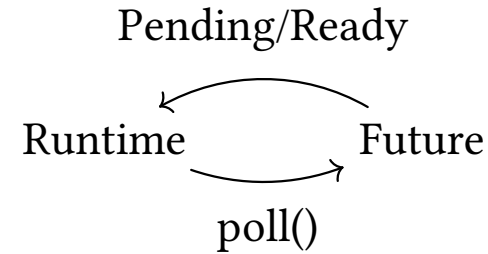
2.2. Comparison with Python

Python asyncio



- Callback-based model
- Register callbacks
- Built-in event loop
- Interpreter overhead

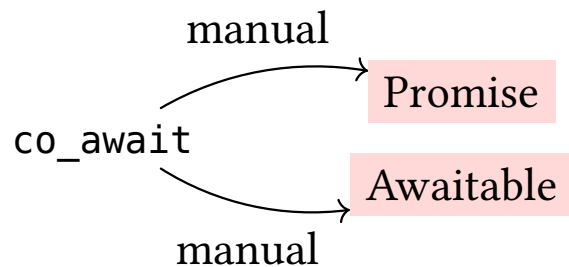
Rust async/await



- Poll-based model
- Runtime polls futures
- External runtimes (tokio, async-std)
- Zero-cost state machines

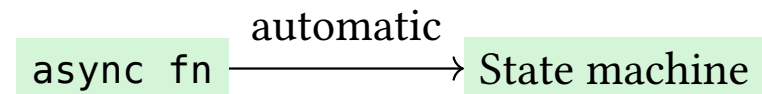
2.3. Comparison with C++

C++20 Coroutines



- Manual promise types
- Manual awaitable objects
- No standard runtime
- **Heap allocation** by default

Rust async/await



- Compiler generates state machines
- Built-in syntax
- Mature ecosystem (tokio)
- **Stack-allocated** by default

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3.1. Await syntax

At a high level, async Rust code looks very much like “normal” sequential code:

```
1  use futures::executor::block_on;
2
3  async fn count_to(count: i32) {
4      for i in 0..count {
5          println!("Count is: {i}!");
6      }
7  }
8
9  async fn async_main(count: i32) {
10     count_to(count).await;
11 }
12
13 fn main() {
14     block_on(async_main(10));
15 }
```

([playground link](#))

3.2. Futures

Future is a trait, implemented by objects that represent an operation that may not be complete yet. A future can be polled, and poll returns a Poll.

```
1  use std::pin::Pin; use std::task::Context;
2
3  pub trait Future {
4      type Output;
5      fn poll(self: Pin<&mut Self>, cx: &mut Context<'_>) -> Poll<Self::Output>;
6  }
7
8  pub enum Poll<T> {
9      Ready(T),
10     Pending,
11 }
```

([playground link](#))

What is the role of Context?

3.2. Futures

Future is a trait, implemented by objects that represent an operation that may not be complete yet. A future can be polled, and poll returns a Poll.

```
1 use std::pin::Pin; use std::task::Context;
2
3 pub trait Future {
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6 }
7
8 pub enum Poll<T> {
9     Ready(T),
10    Pending,
11 }
```

([playground link](#))

What is the role of Context?

Context allows a Future to schedule itself to be polled again when an event such as a timeout occurs.

3.3. JoinHandle

One common type implementing Future is a Tokio JoinHandle:

```
1 use tokio::task;  
2  
3 let join = task::spawn(async {  
4     panic!("something bad happened!")  
5 });  
6  
7 // The returned result indicates that the task failed.  
8 assert!(join.await.is_err());
```

[*\(playground link\)*](#)

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4.1. State machine

Rust transforms async functions into state machines that implement Future.

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Key characteristics:

- Calling an async function constructs and returns a future
- Does not execute immediately (lazy)
- Tracks progress through suspension points

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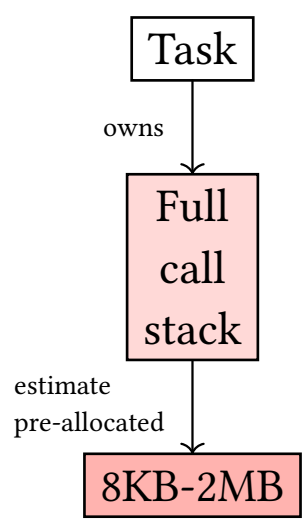
The state machines are allocated on the stack by default. Once allocated, the async runtime never moves the state machine to a different memory location.

Definition

Rust uses **stackless coroutines**: async functions don't have their own call stack. Instead, they compile to state machines that store only the data needed at suspension points, enabling zero-cost abstractions.

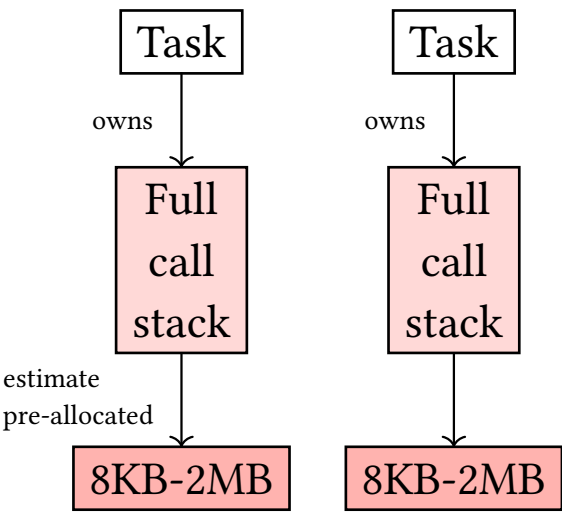
4.2. Stackless vs Stackful

Stackful (Go, goroutines)



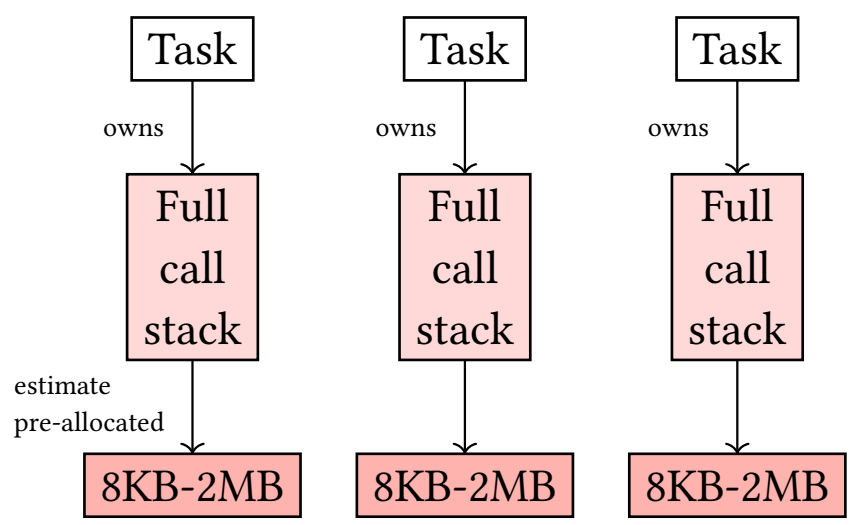
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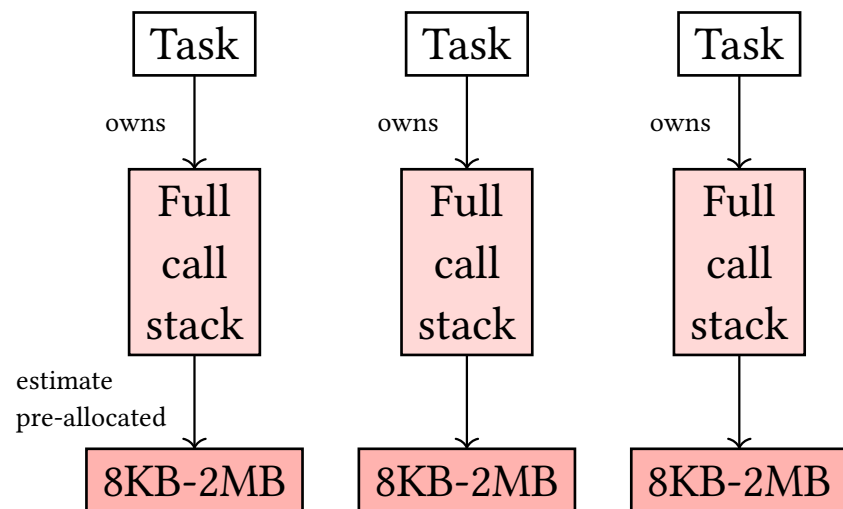
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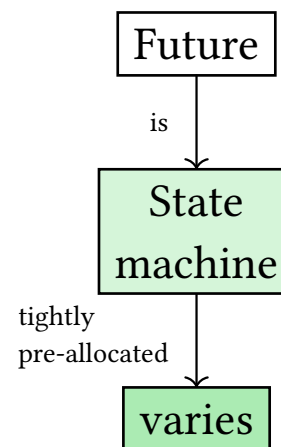
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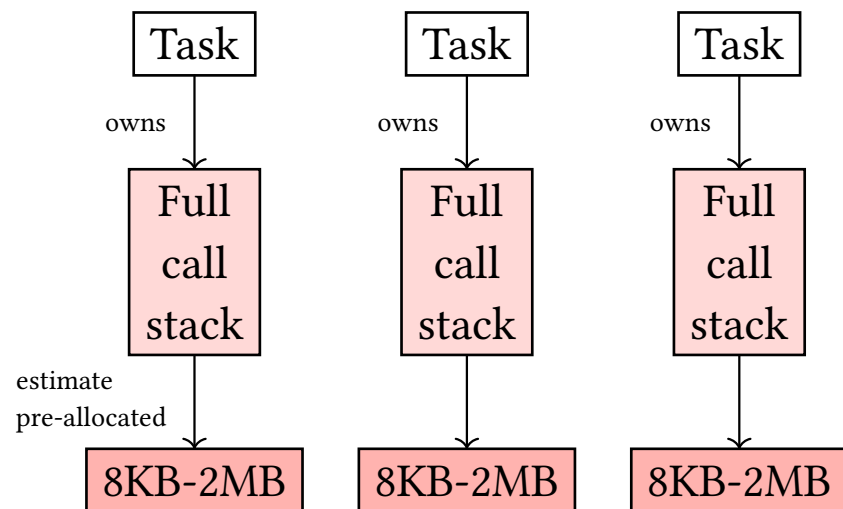
10K tasks = 80MB-20GB

Stackless (Rust async)



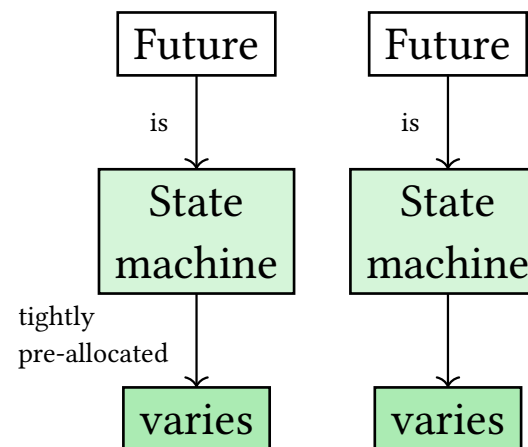
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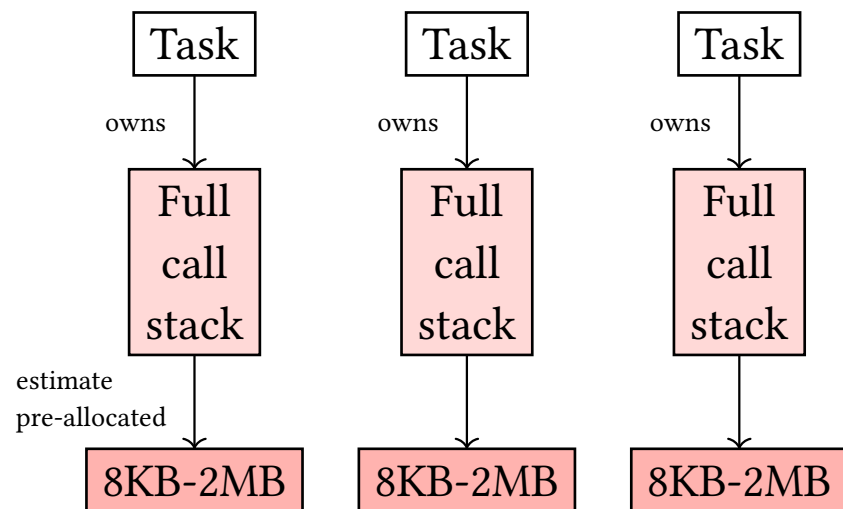
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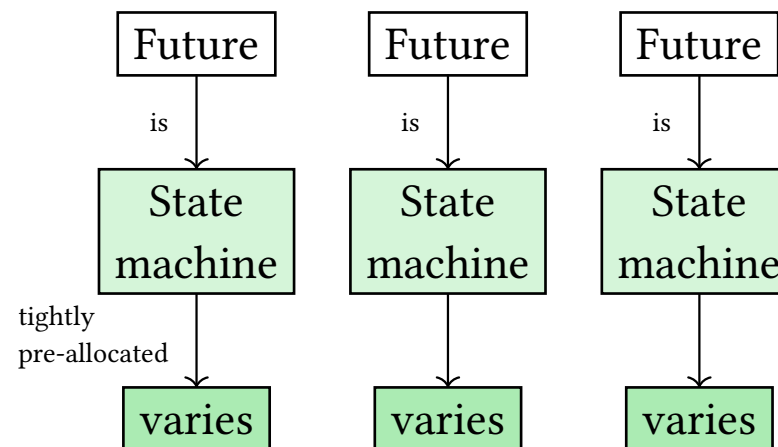
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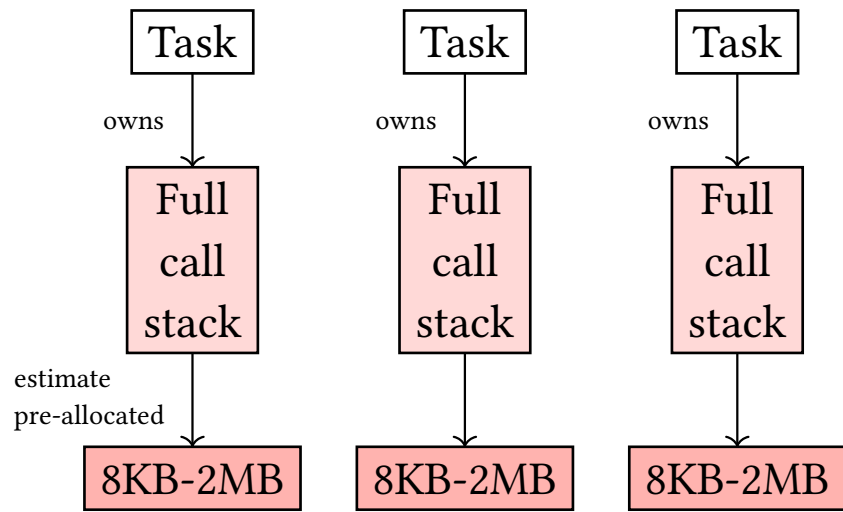
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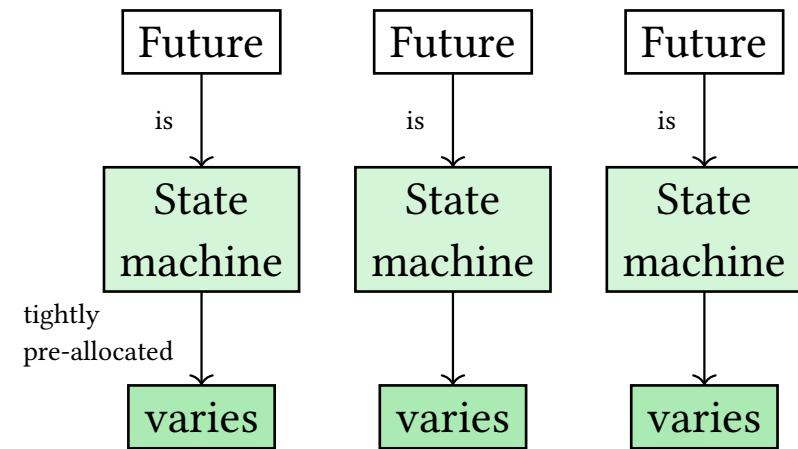
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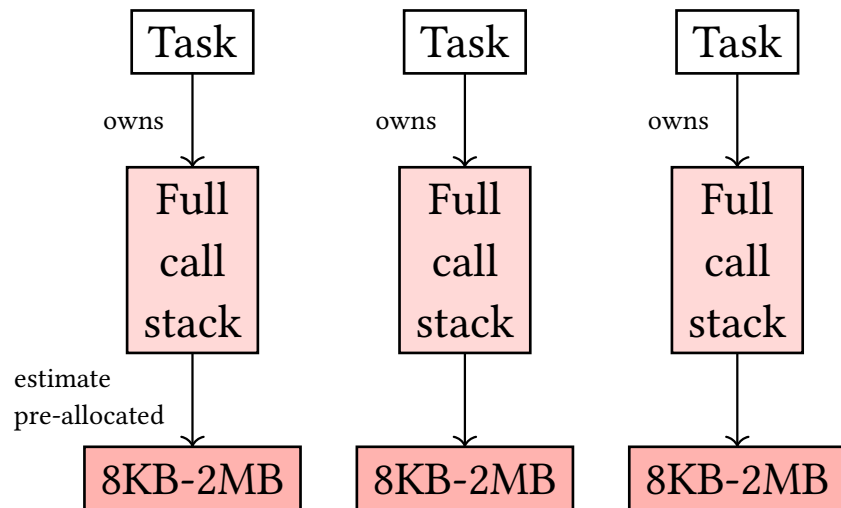
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10K tasks = 1MB

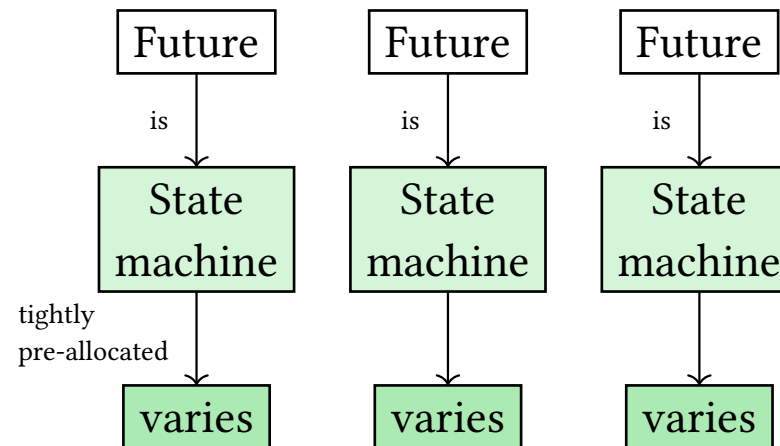
4.2. Stackless vs Stackful

Stackful (Go, goroutines)



10K tasks = 80MB-20GB

Stackless (Rust async)



10K tasks = 1MB

Key difference: Stackful tasks allocate a fixed-size stack (pre-allocated for deep call chains), while stackless state machines allocate only the exact space needed for live variables at each suspension point.

4.3. Transformation

Consider this async function:

```
1 async fn two_d10(modifier: u32) -> u32 {  
2     let first_roll = roll_d10().await;  
3     let second_roll = roll_d10().await;  
4     first_roll + second_roll + modifier  
5 }
```

[\(playground link\)](#)

4.3. Transformation

Consider this async function:

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4     first_roll + second_roll + modifier  
5 }
```

[\(playground link\)](#)

The compiler generates an enum that:

- Tracks each suspension point (each `.await`)
- Stores all live variables needed at that state
- Implements the Future trait

4.4. State machine structure

The generated enum looks conceptually like this:

```
1 enum TwoDiceFuture {  
2     Start { modifier: u32 },  
3     FirstRoll { modifier: u32, first_roll: impl Future<Output = u32> },  
4     SecondRoll { modifier: u32, first_result: u32, second_roll: impl Future<Output =  
5         u32> },  
6     Done,  
7 }
```

[\(playground link\)](#)

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5         u32> },  
6     Done,  
}
```

[\(playground link\)](#)

Each state stores:

- The variables still needed
- The future being awaited (if any)

4.5. Recursion

Deeply nested async functions create large compiler-generated Future types.

Each function's Future contains its callees' Futures.

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Each function's Future contains its callees' Futures.

Recursive async functions require boxing to avoid infinite type sizes:

```
1 async fn count_to(n: u32) {  
2     if n > 0 {  
3         Box::pin(count_to(n - 1)).await;  
4         println!("{n}");  
5     }  
6 }  
7  
8 Box::pin(count_to(n - 1)).await;
```

([playground link](#))

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([playground link](#))

This adds heap allocation overhead but makes recursion possible.

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5.1. Runtimes

A runtime provides support for:

- performing operations asynchronously (a **reactor**)
- and is responsible for executing futures (an **executor**).

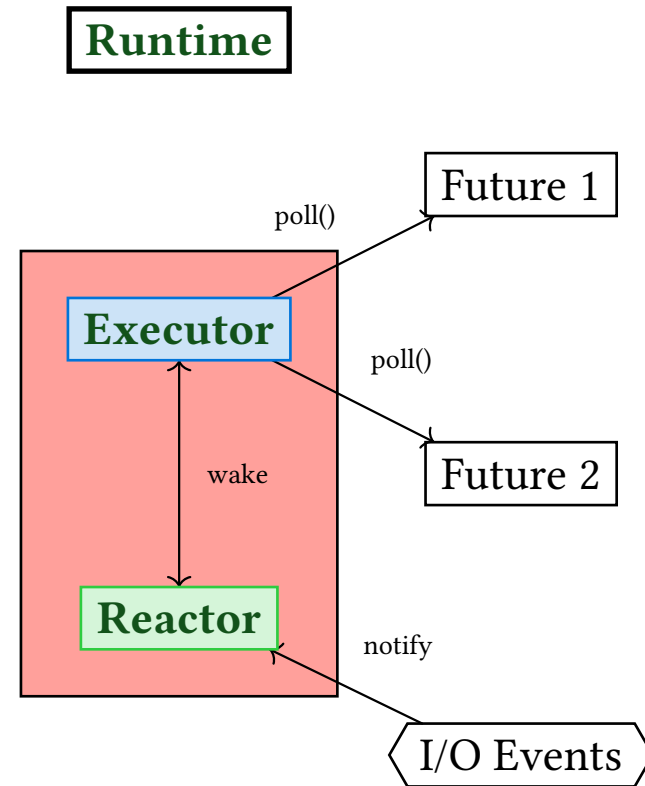
5.1. Runtimes

A runtime provides support for:

- performing operations asynchronously (a **reactor**)
- and is responsible for executing futures (an **executor**).

Rust does not have a “built-in” runtime, but several options are available:

- futures offers a bare bones executor ThreadPool
- tokio offers a Runtime with reactor included (supplies “time” etc.)



```
1  use tokio::time;
2  async fn count_to(count: i32) {
3      for i in 0..count {
4          println!("Count in task: {i}!");
5          time::sleep(time::Duration::from_millis(5)).await;
6      }
7  }
8  #[tokio::main]
9  async fn main() {
10     tokio::spawn(count_to(10));
11     for i in 0..5 {
12         println!("Main task: {i}");
13         time::sleep(time::Duration::from_millis(5)).await;
14     }
15 }
```

[\(playground link\)](#)

Async tasks are also aborted if not awaited in main.

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6.1. Tasks

6. Common async datatypes and functions

A task is a top-level future spawned into the runtime. In multi-threaded runtimes, tasks must be `Send + 'static` and may be moved between worker threads by the work-stealing scheduler.

```
1  #[tokio::main]
2  async fn main() -> io::Result<()> {
3      let listener = TcpListener::bind("127.0.0.1:0").await?;
4
5      loop {
6          let (mut socket, addr) = listener.accept().await?;
7
8          tokio::spawn(async move {
9              // Handle connection
10          });
11      }
12 }
```

[*\(playground link\)*](#)

6.1. Tasks

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10          });
11      }
12 }
```

[\(playground link\)](#)

Warning

Although we use tokio in this section, the same functionality is available in most async runtimes.

Inside `tokio::spawn`:

```
1 socket.write_all(b"Who are you?\n").await?;  
2  
3 let mut buf = vec![0; 1024];  
4 let name_size = socket.read(&mut buf).await?;  
5 let name = std::str::from_utf8(&buf[..name_size])?.trim();  
6  
7 socket.write_all(format!("Thanks, {name}!\n").as_bytes()).await?;    (playground link)
```

Tokio Runtime

Tokio Runtime

Executor

Tokio Runtime

Executor

Thread 1

Thread 2

Thread 3

Tokio Runtime

Executor

Reactor

Thread 1

Thread 2

Thread 3

Tokio Runtime

Executor

Reactor

Thread 1

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Task 1

Task 2

Tokio Runtime

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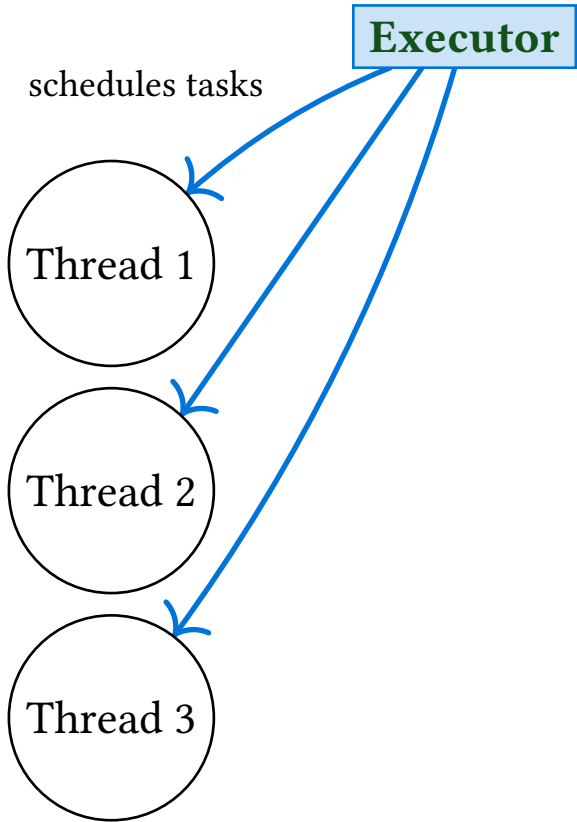
Thread 3

Task 1

Task 2

I/O Events

Tokio Runtime



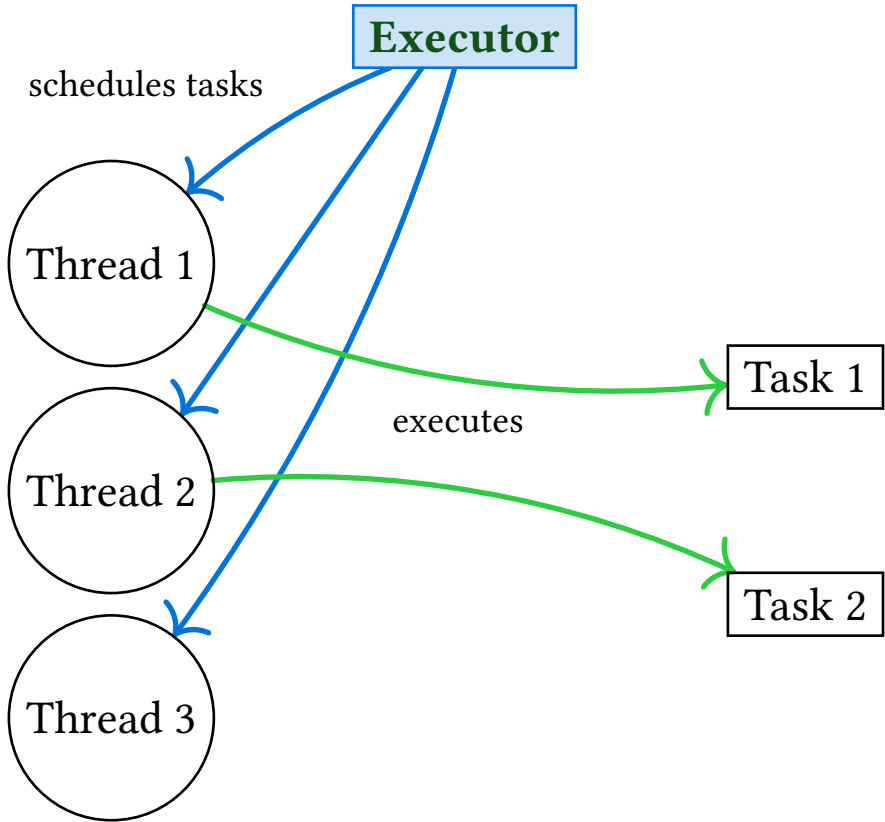
Task 1

Task 2

Reactor

I/O Events

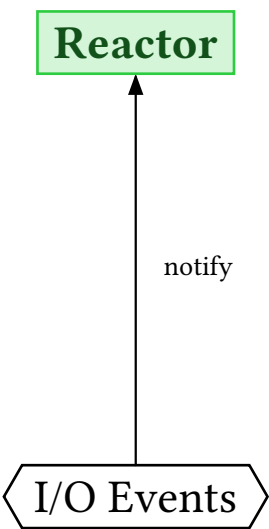
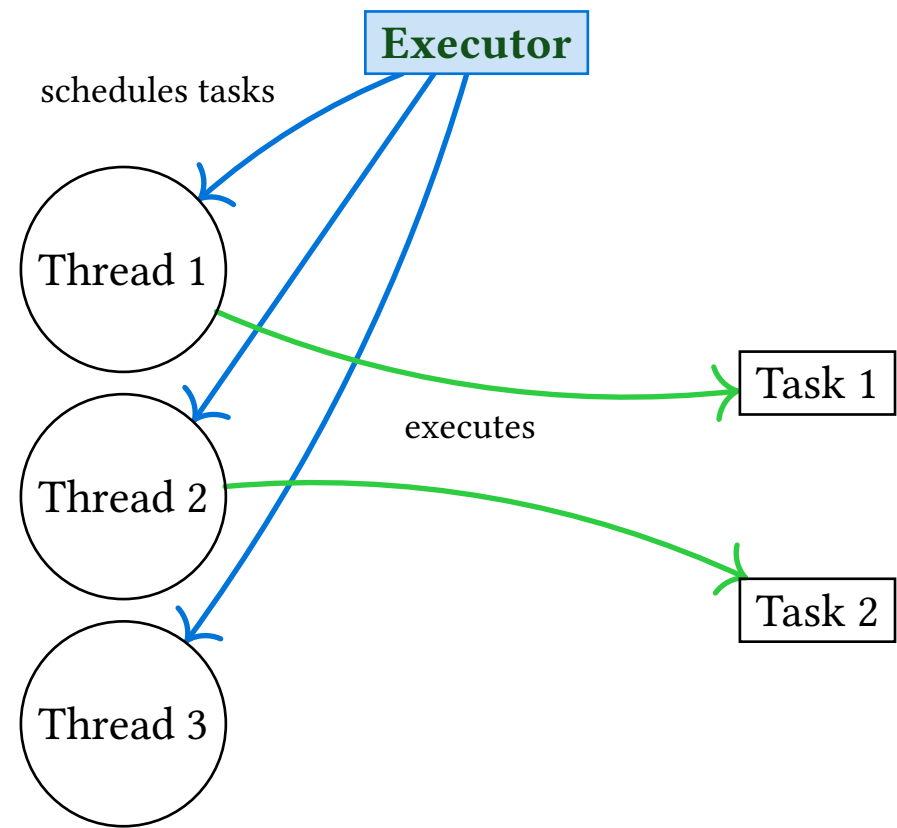
Tokio Runtime

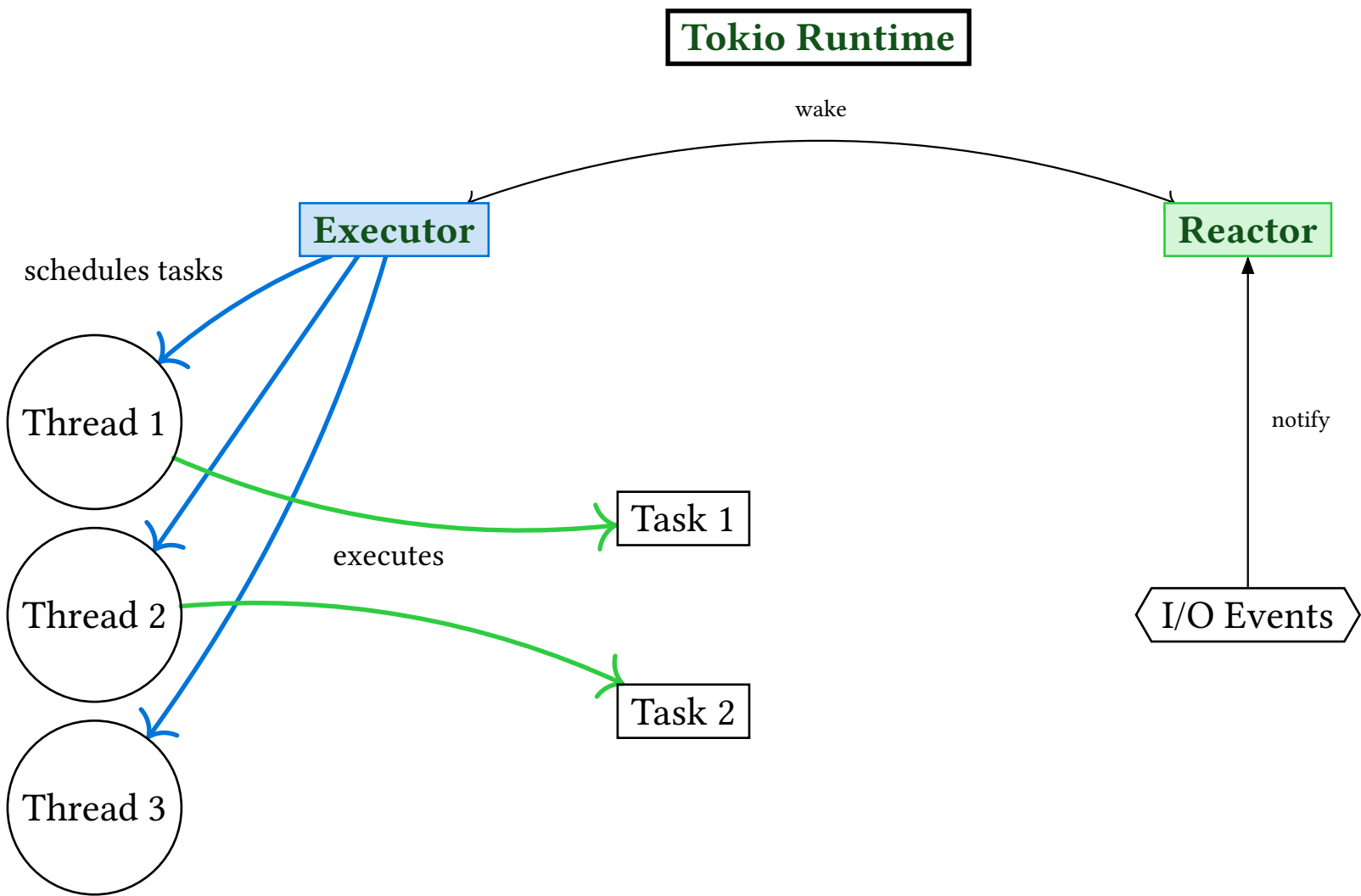


Reactor

I/O Events

Tokio Runtime





Info

An async block like `async move {}` is like a scope block that may contain awaits.

Refactor `session7/examples/tasks.rs`:

- Create async helper function from async block
- Improve error handling

```
1 async fn ping_handler(mut input: mpsc::Receiver<()>) {
2     let mut count: usize = 0;
3     while let Some(_) = input.recv().await {
4         count += 1;
5         println!("Received {count} pings so far.");
6     }
7 }
```

([playground link](#))

```
1 async fn ping_handler(mut input: mpsc::Receiver<()>) {
2     let mut count: usize = 0;
3     while let Some(_) = input.recv().await {
4         count += 1;
5         println!("Received {count} pings so far.");
6     }
7 }
```

[\(playground link\)](#)

```
1 async fn main() {
2     let (sender, receiver) = mpsc::channel(32);
3     let handler_task = tokio::spawn(ping_handler(receiver));
4     for i in 0..10 {
5         sender.send(()).await.expect("Failed to send");
6     }
7     drop(sender);
8     handler_task.await.expect("Handler failed");
9 }
```

[\(playground link\)](#)

We might need to do the same async function multiple times:

```
1 async fn size_of_page(url: &str) -> Result<usize> {  
2     let resp = reqwest::get(url).await?;  
3     Ok(resp.text().await?.len())  
4 }
```

[\(playground link\)](#)

How can we await a list of futures concurrently?

6.4. Joining

We might need to do the same async function multiple times:

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[\(playground link\)](#)

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We use the `join_all` function.

6.4. Joining

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[\(playground link\)](#)

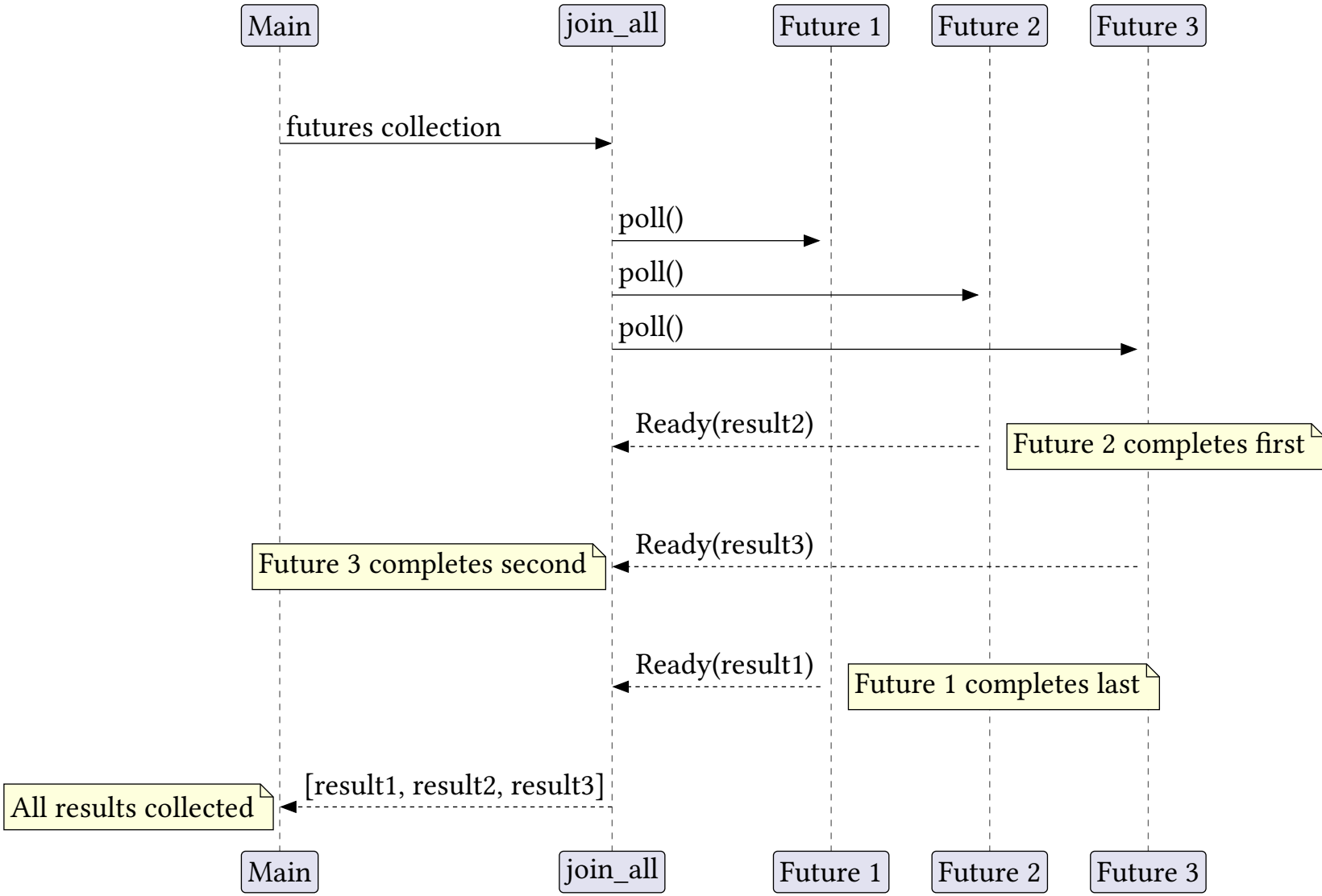
How can we await a list of futures concurrently?

We use the `join_all` function.

```
1 async fn main() {  
2     let urls = ["https://google.com", "https://httpbin.org/ip", /*...*/];  
3     let futures = urls.into_iter().map(size_of_page);  
4     let results = future::join_all(futures).await;  
5     println!("{results:?}");  
6 }
```

[\(playground link\)](#)

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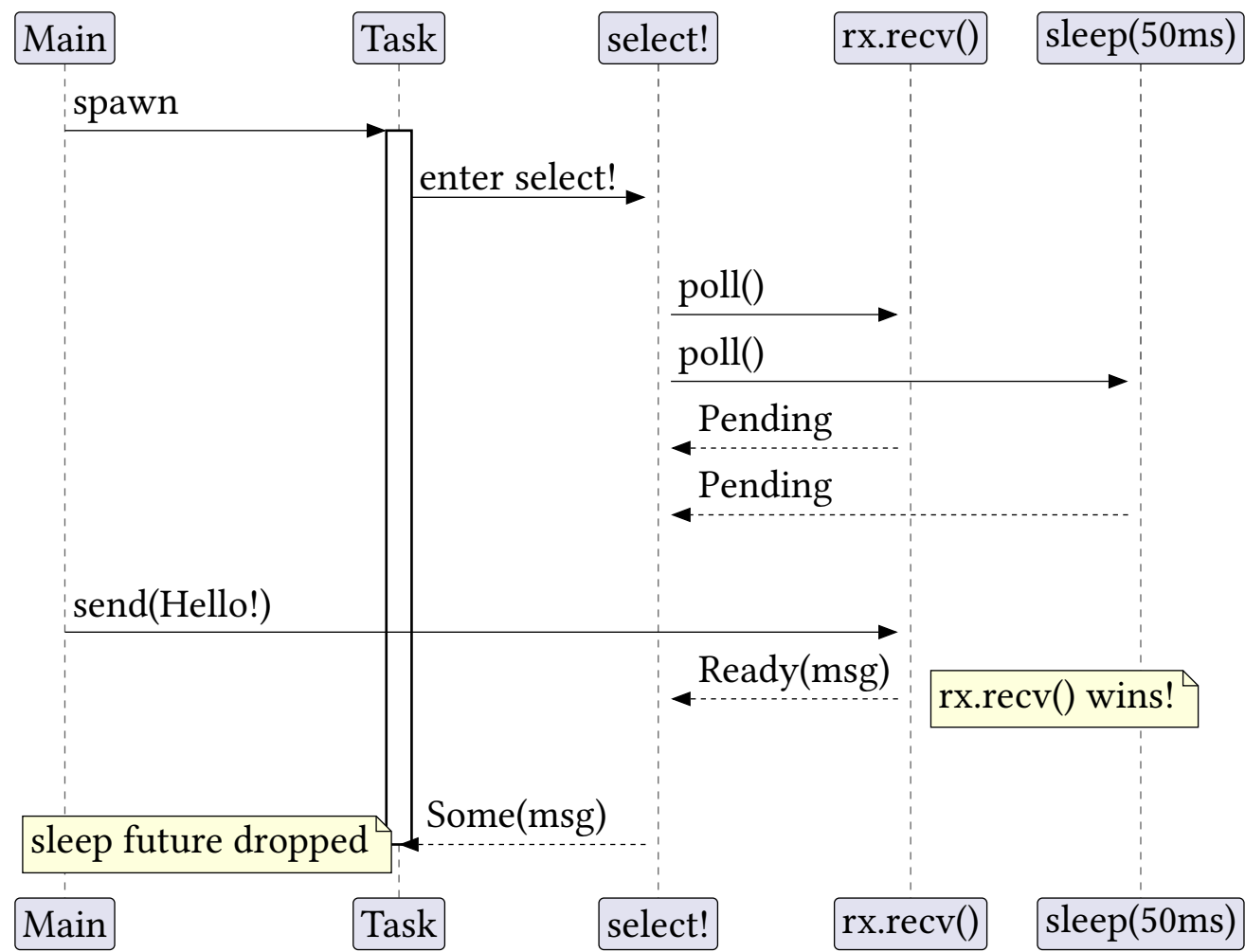
6.5. Select

Waits until any of a set of futures is ready. Similar to `Promise.race` or Python's `asyncio.wait()`.

```
1  #[tokio::main]
2  async fn main() {
3      let (tx, mut rx) = mpsc::channel(32);
4
5      tokio::spawn(async move {
6          tokio::select! {
7              Some(msg) = rx.recv() => println!("got: {msg}"),
8              _ = sleep(Duration::from_millis(50)) => println!("timeout"),
9          };
10     });
11
12     sleep(Duration::from_millis(10)).await;
13     tx.send(String::from("Hello!")).await?;
14     // ...
15 }
```

([playground link](#))

6.5. Select



Streams are runtime-agnostic asynchronous iterators:

```
1 use futures::stream::{self, StreamExt};
2
3 let stream = stream::iter(vec!['a', 'b', 'c']);
4
5 let mut stream = stream.enumerate();
6
7 assert_eq!(stream.next().await, Some((0, 'a')));
8 assert_eq!(stream.next().await, Some((1, 'b')));
9 assert_eq!(stream.next().await, Some((2, 'c')));
10 assert_eq!(stream.next().await, None);
```

[\(playground link\)](#)

See my project <https://github.com/wvhulle/clone-stream>

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7.1. Blocking executor

CPU blocking tasks will block the executor and prevent other tasks from being executed.

```
1  async fn sleep_ms(start: &Instant, id: u64, duration_ms: u64) {
2      std::thread::sleep(std::time::Duration::from_millis(duration_ms));
3      println!("future {id} finished after {}ms", start.elapsed().as_millis());
4  }
5
6  #[tokio::main(flavor = "current_thread")]
7  async fn main() {
8      let start = Instant::now();
9      let futures = (1..=10).map(|t| sleep_ms(&start, t, t * 10));
10     join_all(futures).await;
11 }
```

([playground link](#))

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([playground link](#))

What is a workaround?

Use async methods: `tokio::time::sleep` instead of `std::thread::sleep`

7.2. Async for limited parallelism

Warning

Async tasks run on a limited thread pool and don't provide true parallelism for CPU-bound work.

Async is designed for I/O concurrency, not CPU parallelism:

- Spawning 10,000 async tasks doesn't create 10,000 workers
- Tasks execute sequentially on the available thread pool
- CPU-bound work sees no benefit from async spawning

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- Tasks execute sequentially on the available thread pool
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Recommendation: If there's no real I/O waiting:

- Use a small number of OS threads instead (`std::thread` or `rayon`)
- Match thread count to CPU cores for optimal performance
- Reserve async for coordinating I/O operations

7.3. Avoid spawn-ception

Warning

Spawning tasks from within spawned tasks creates hard-to-debug complexity.

Common anti-pattern:

```
1 tokio::spawn(async {  
2     // Some work  
3     tokio::spawn(async { // Nested spawn!  
4         // More work  
5         tokio::spawn(async { /* ... */ })  
6     })  
7 })
```

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[\(playground link\)](#)

Recommendation: Spawn once at application boundaries:

- Spawn tasks at the top level (e.g., per incoming request)
- Use regular async functions and `.await` for sequential steps
- Only spawn when you need true concurrent execution

7.4. Deadlocks: Coffman Conditions

A deadlock occurs when four conditions are met simultaneously (Coffman conditions):

Condition	Description
Mutual exclusion	Resources cannot be shared (only one thread at a time)
Hold and wait	Threads hold resources while waiting for others
No preemption	Resources cannot be forcibly taken from threads
Circular wait	Threads form a cycle waiting for each other's resources

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Circular wait	Threads form a cycle waiting for each other's resources

To prevent deadlock, break at least one condition:

- **Lock ordering:** Always acquire locks in the same order (breaks circular wait)
- **Timeout:** Release locks if waiting too long (breaks hold and wait)
- **Try-lock:** Use `try_lock()` instead of `lock()` (breaks hold and wait)

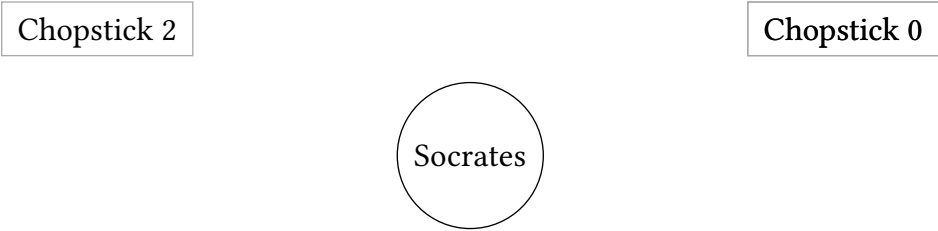
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8.1. Context

Simplified: 3 philosophers sitting around a table, 3 chopsticks between them.

Rules:

- Each philosopher needs 2 chopsticks (left and right) to eat
- Eating takes time (must hold both chopsticks during eating)
- All philosophers want to eat repeatedly

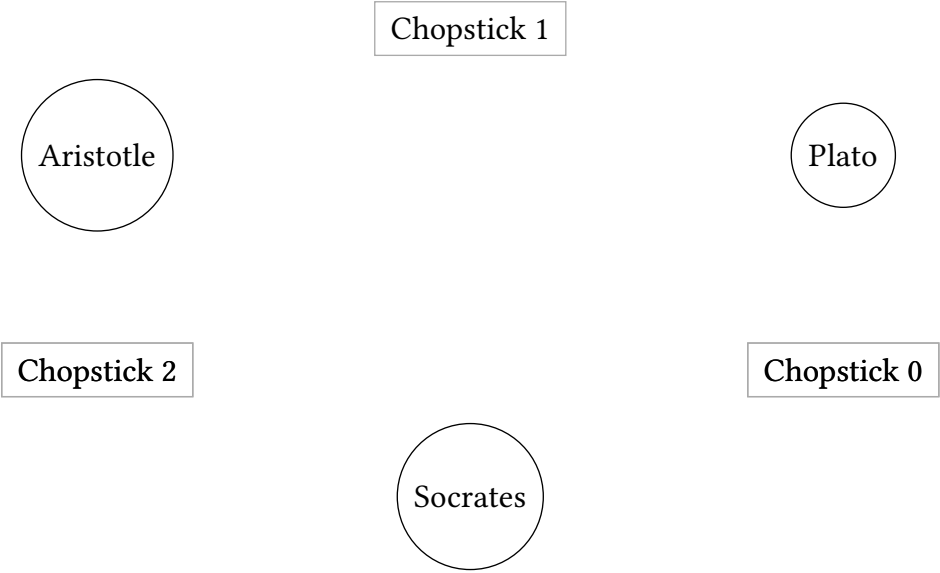


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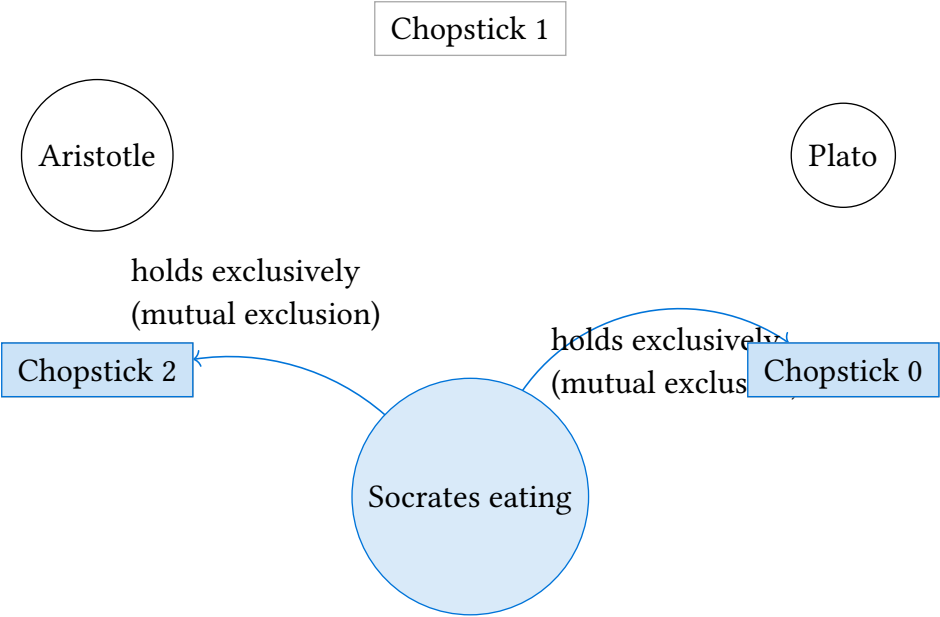


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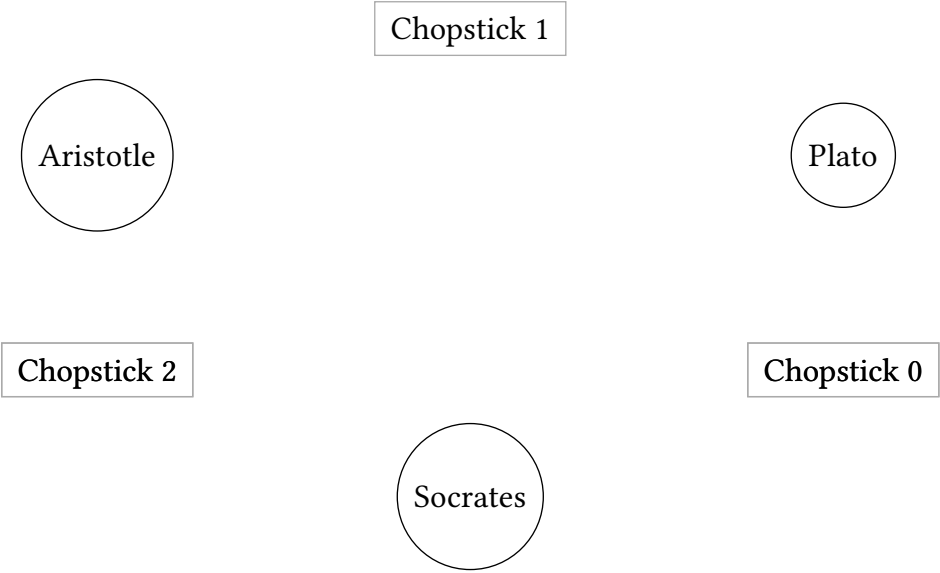


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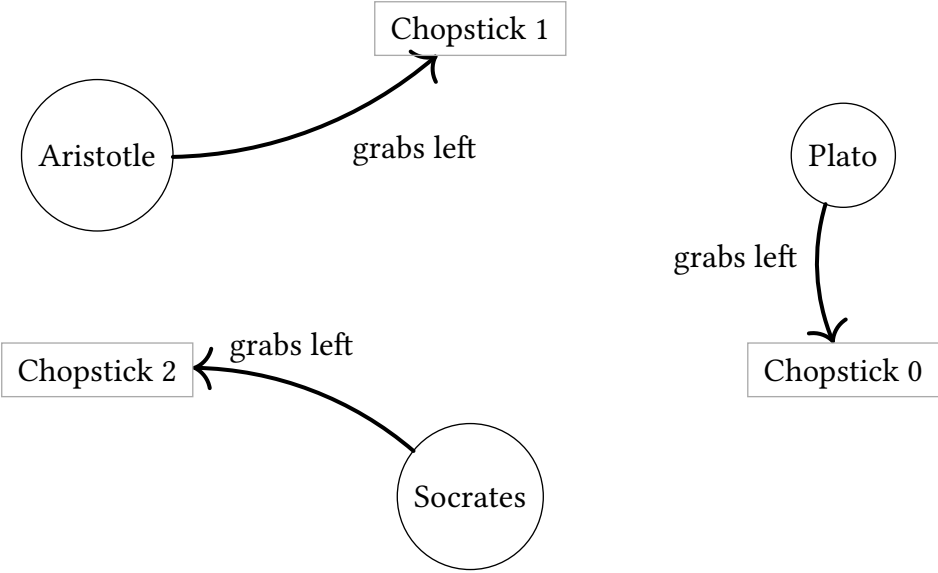


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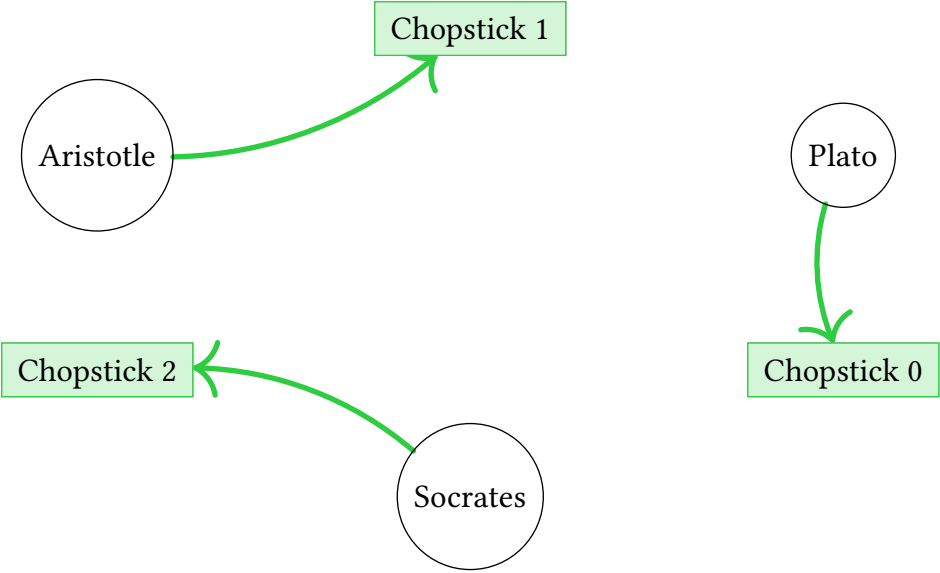


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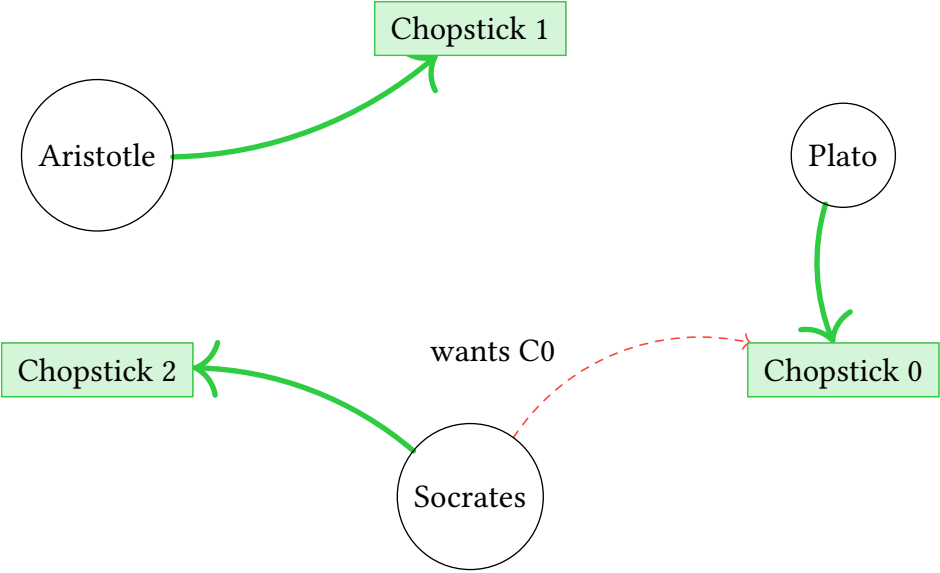


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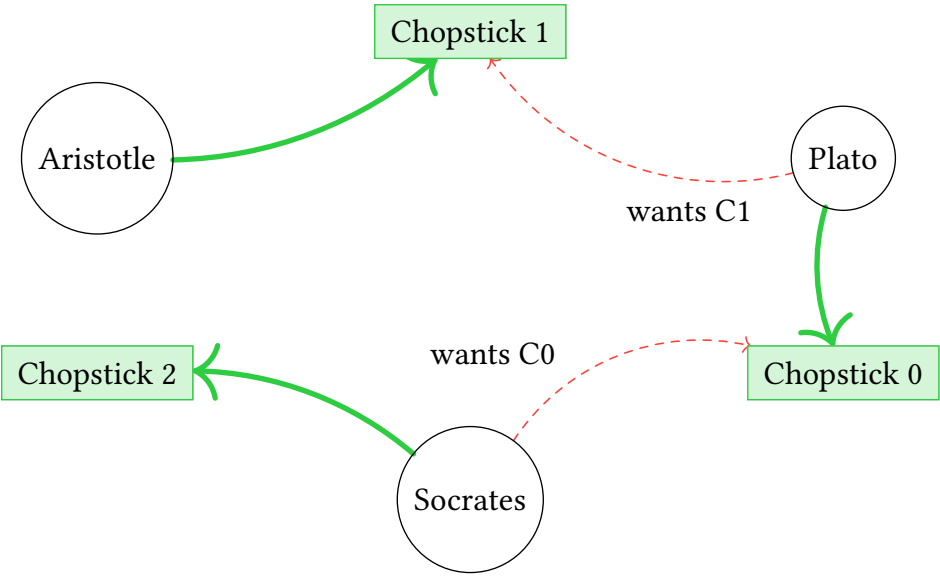


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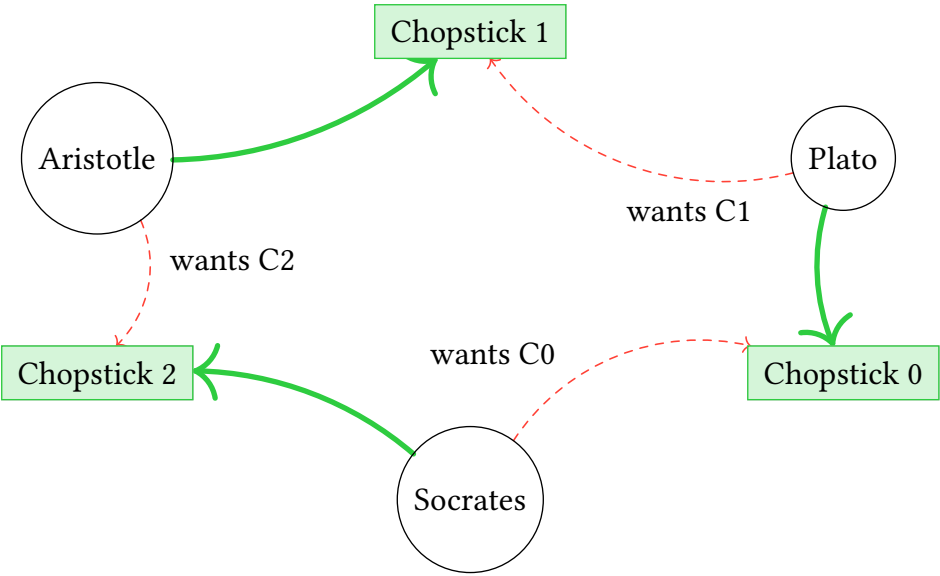


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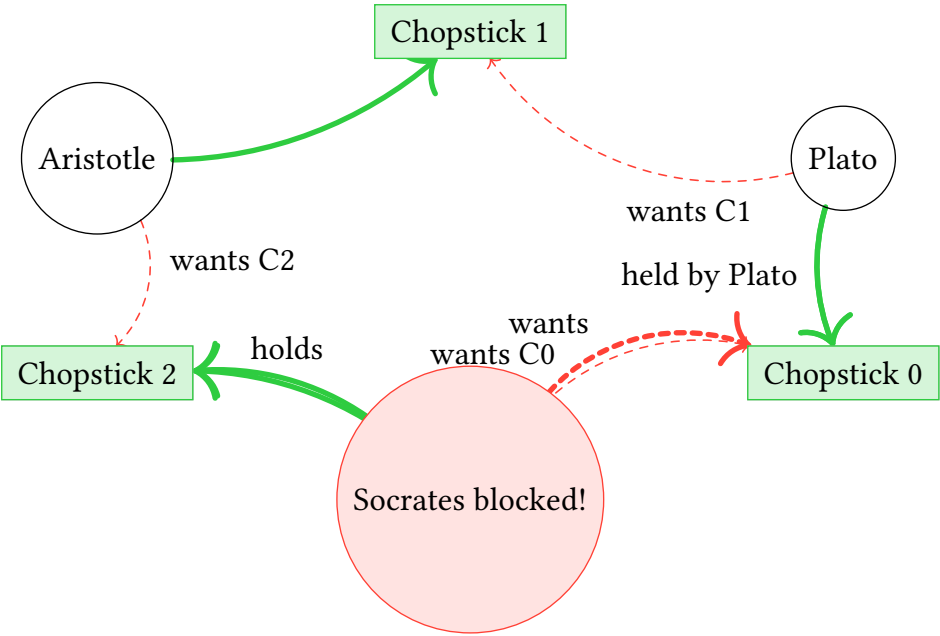


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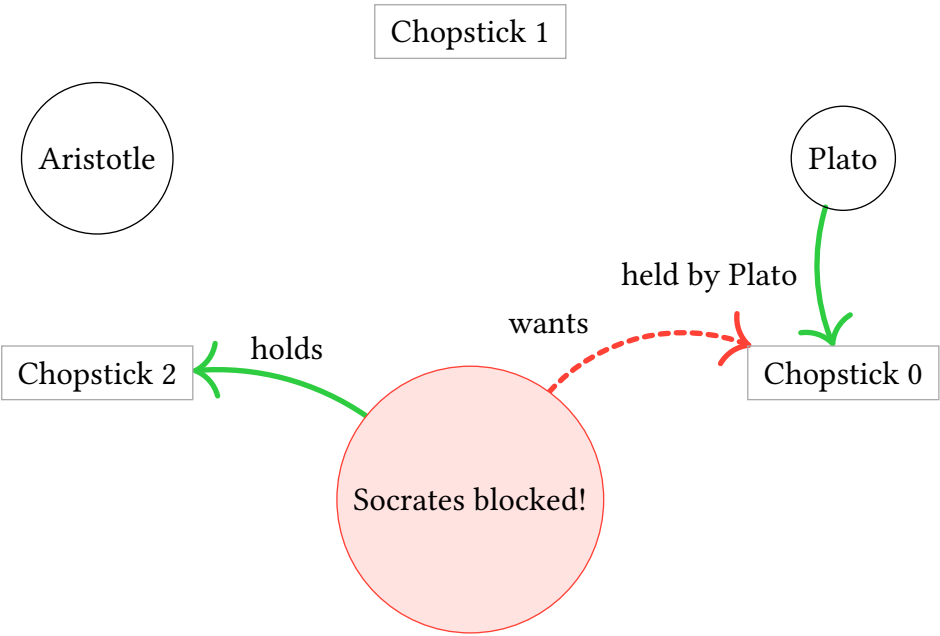


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Legend:

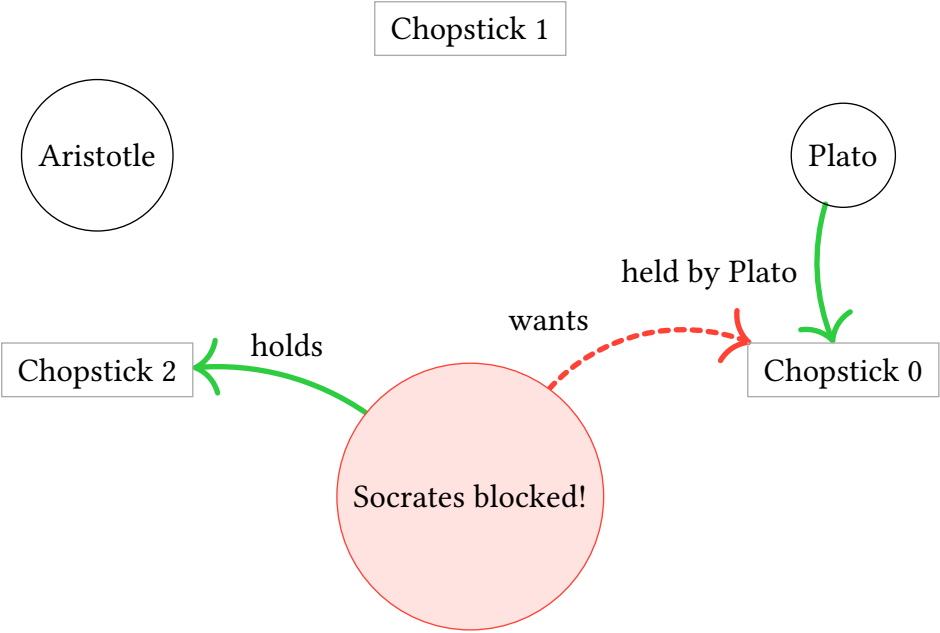
- Trying to grab
- Holding successfully
- - - Wants but blocked

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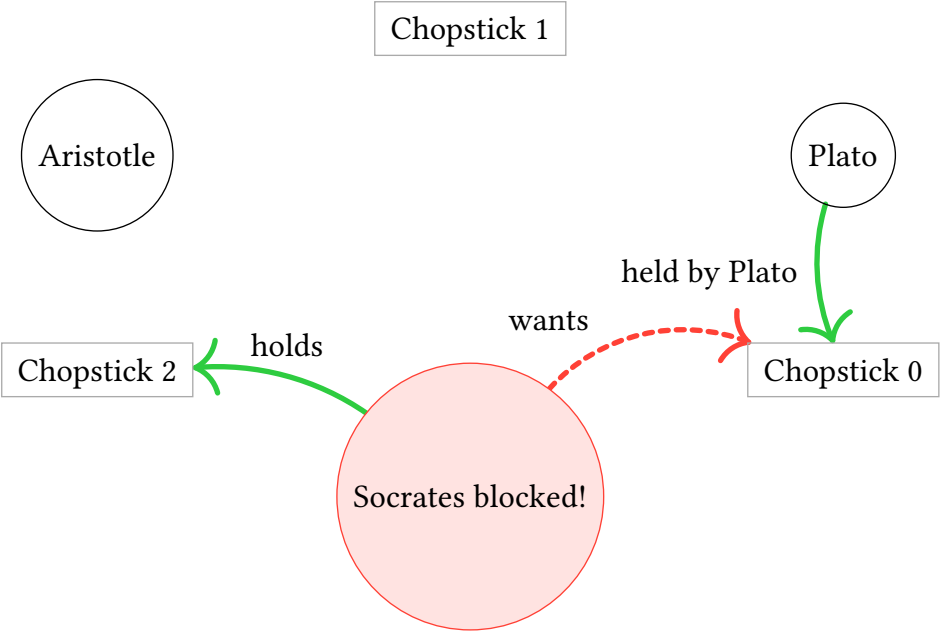
- Holds C2 (green)
- Wants C0 (red dashed)
- But C0 is held by Plato!

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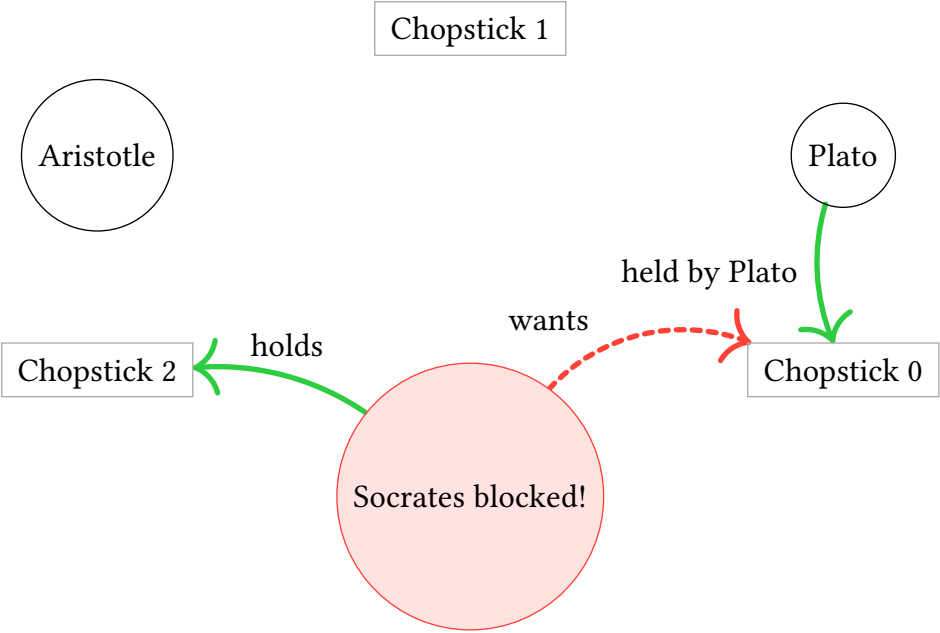
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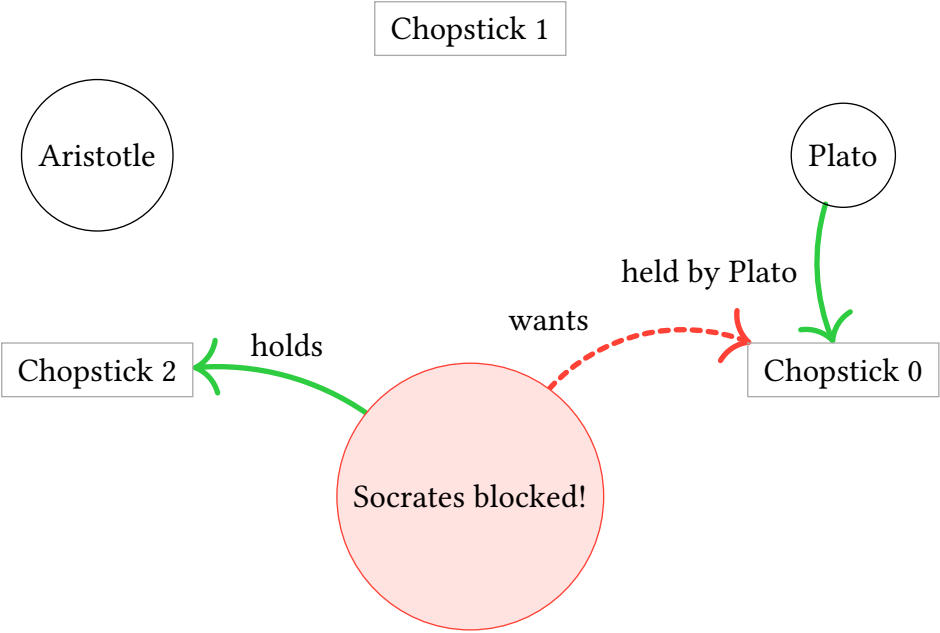
Which Coffman condition should we break?

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Same for Plato and Aristotle → circular wait → **Deadlock!**

Which Coffman condition should we break?

Circular wait: change lock order for one philosopher.

8.2. Assignment

Solve the “Dining Philosophers” problem with async tasks instead of threads.

- Revisit the standard thread-based version in `session6/examples/philosophers.rs`
- Complete exercise code in `session7/examples/philosophers.rs`.

Solutions are provided as `*-solution.rs` files.

8.3. Additional bonus

8. Exercise: async philosophers

Do we still need the asymmetry (the last philosopher switching hands)?

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Yes, because all philosophers are still working in parallel in separate tasks, so we may have a deadlock.

How to solve this problem with only a single async task? Do we still have deadlocks

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No, since philosophers access sequentially.